



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Building Multi Agent Applications with Gemini ADK

Course Code: C829

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	4 October 2026	First release of the non-WSQ Lesson Plan for Building Multi Agent Applications with Gemini ADK (C829). Two days, 7.5 instructional hours per day, 9:30 am – 5:30 pm.	Dr Alfred Ang

TABLE OF CONTENTS

Course Information	4
Learning Outcomes	4
Course Schedule	4
Day 1 — Agentic AI foundations with Gemini ADK — first agents, tools, sessions and multi-agent handoff.....	4
Day 2 — Multi-agent systems, agentic RAG and shipping an agent app with Streamlit.....	5
Lab Reference (aligned to course topics)	6

Course Information

Course Title	Building Multi Agent Applications with Gemini ADK
Course Reference	C829
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 7.5 instructional hours per day (15 hours)
Daily Timing	9:30 am – 5:30 pm (30-minute lunch; refreshments during activity time)
Mode	Instructor-led, hands-on agent-development labs using Python, the Google Agent Development Kit and Gemini
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Analyze the range of LLM applications using Generative AI (GAI) and identify their industrial use cases.
- LO2: Establish Google Gemini GAI designs and assess improvements on engineering processes.
- LO3: Develop LLM applications and assess its feasibility.
- LO4: Evaluate the performance effectiveness of Retrieval Augmented Generation (RAG).

Course Schedule

Day 1 — Agentic AI foundations with Gemini ADK — first agents, tools, sessions and multi-agent handoff

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, learning outcomes and ground rules
10:00–11:15	75 min	Topic 1 — Overview of Agentic AI in Gemini ADK: what an LLM is, LLM use cases across industries, agentic AI, the Gemini model family and ADK architecture (concepts + demo)
11:15–12:45	90 min	Hands-on: Lab 1: Set Up the Gemini ADK Environment and Get an API Key; Lab 2: Build Your First ADK Agent — A Retail Banking Assistant
12:45–13:15	30 min	Lunch break
13:15–15:00	105 min	Hands-on: Lab 3: Give an Agent Tools — Live Weather

		and Web Search; Lab 4: Swap the Model — Running an ADK Agent on a Non-Gemini LLM
15:00–15:45	45 min	Topic 2 — Build A Multi Agent App with Gemini ADK: sessions and state, the Runner, the event loop, multi-agent handoff and the coordinator/dispatcher pattern (concepts + demo)
15:45–17:15	90 min	Hands-on: Lab 5: Give an Agent Memory — Sessions, State and the Runner; Lab 6: Inspect the Agent Loop — Events, Tool Calls and Final Responses; Lab 7: Multi-Agent Handoff — A Blog Content Production Team; Lab 8: Coordinator / Dispatcher — A Customer Support Desk
17:15–17:30	15 min	Day 1 recap and Q&A

Day total: 450 minutes — 7.5 instructional hours (9:30am – 5:30pm).

Day 2 — Multi-agent systems, agentic RAG and shipping an agent app with Streamlit

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap
9:45–12:00	135 min	Topic 2 continues — workflow agents (Sequential, Parallel, Loop and Agent-as-a-Tool), guardrails, structured output and MCP (concepts). Hands-on: Lab 9: Workflow Agents — Sequential, Parallel, Loop and Agent-as-a-Tool; Lab 10: Add a Guardrail — Blocking Unsafe Requests with a Callback
12:00–12:30	30 min	Lunch break
12:30–14:40	130 min	Hands-on: Lab 11: Structured Output — Forcing Valid JSON with Pydantic; Lab 12: Connect External Tools with MCP — StreamableHTTP and SSE
14:40–16:30	110 min	Topic 3 — Build Agentic AI RAG in Gemini ADK: the RAG pipeline, chunking, embeddings, vector stores and retrieval (concepts). Hands-on: Lab 13: Load, Split and Embed Documents into a Vector Store; Lab 14: Build the Agentic RAG Agent — Retrieval as a Tool; Lab 15: Evaluate RAG Performance — Retrieval Quality and Groundedness
16:30–17:15	45 min	Topic 4 — Agentic AI App with Gemini ADK and Streamlit

		(concepts). Hands-on: Lab 16: Declarative Agents — Configuring a Multi-Agent System in YAML; Lab 17: Ship the Agent as a Web App with Streamlit; Lab 18: Capstone — Build Your Own Multi-Agent Application
17:15–17:30	15 min	Course recap, Q&A and course feedback

Day total: 450 minutes — 7.5 instructional hours (9:30am – 5:30pm).

Lab Reference (aligned to course topics)

Topic / Skill area	Weighting	Labs
Topic 01: Overview of Agentic AI in Gemini ADK	25%	Lab 1, Lab 2, Lab 3, Lab 4
Topic 02: Build A Multi Agent App with Gemini ADK	35%	Lab 5, Lab 6, Lab 7, Lab 8, Lab 9, Lab 10, Lab 11, Lab 12
Topic 03: Build Agentic AI RAG in Gemini ADK	25%	Lab 13, Lab 14, Lab 15
Topic 04: Build an Agentic AI App with Gemini Agent ADK and Streamlit	15%	Lab 16, Lab 17, Lab 18